25901672 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCEVS100 - ECOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **1** marks

1. What is the significance of age distribution in population structure? [K] / [CO1]
2. What is the definition of carrying capacity? [K] / [CO1]
3. What is the term for a group of interacting species in a specific area? [K] / [CO2]
4. What is community density in ecology? [K] / [CO2]
5. Name any two important characters of ecological niche. [K] / [CO2]
6. Which is the community structure characterised by dominant species? [A] / [CO2]
7. What are the important abiotic characters of a community ecology? [U] / [CO2]
8. Give an example for climax community in succession. [U] / [CO3]
9. Differentiate parasitism and allelopathy. [U] / [CO3]
10. Describe secondary succession. [A] / [CO3]

11. Categorize sedge meadow and woodland stage of succession. [An] / [CO5]

12. Illustrate the significance of ecological interactions. [An] / [CO5]

[1x10 = 10]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

13. Describe the different branches of ecology. [K] / [CO1]

14. Differentiate between density-dependent and density-independent factors affecting population dispersal with examples. [U] / [CO4]

15. How do natality and mortality rates influence population growth? [U] / [CO4]

16. Briefly explain how species dominance influences floristic composition? [A] / [CO2]

17. Describe the concept of a community gradient and explain its significance in ecology. [U] / [CO2]

18. Describe the monoclimax and polyclimax theories of climax communities. [U] / [CO3]

19. Correlate different process of ecological succession. [An] / [CO5]

20. Describe the stages of xerosere succession. [U] / [CO5]

[5x6 = 30]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

21. Explain the advantages and limitations of both transect and quadrat methods in community ecology research. [U] / [CO2]

22. Describe the mutualism and commensalism with suitable examples. [U] / [CO3]

[10x1 = 10]